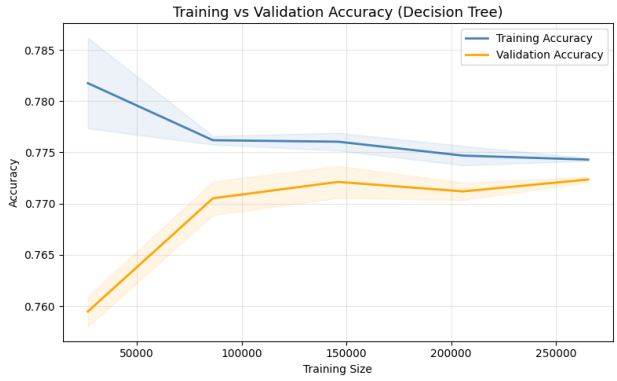
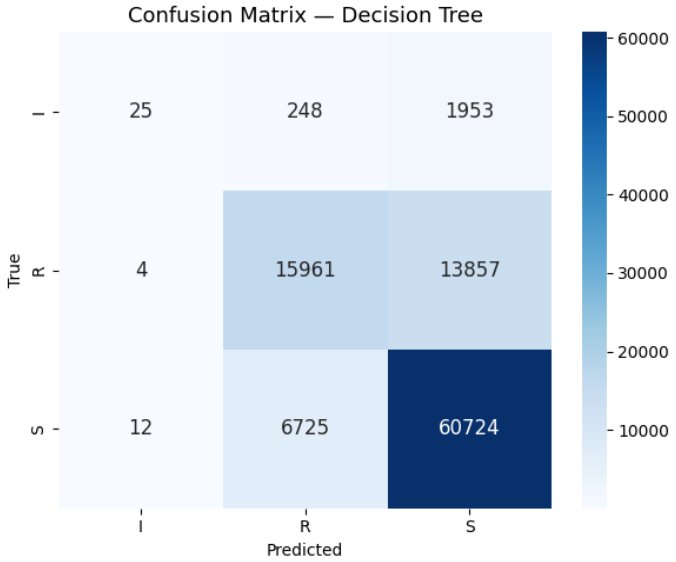
Decision Tree

ชื่อเชื้อ+ชื่อยา + Specimen + เพศ + อายุ + หอผู้ป่วย



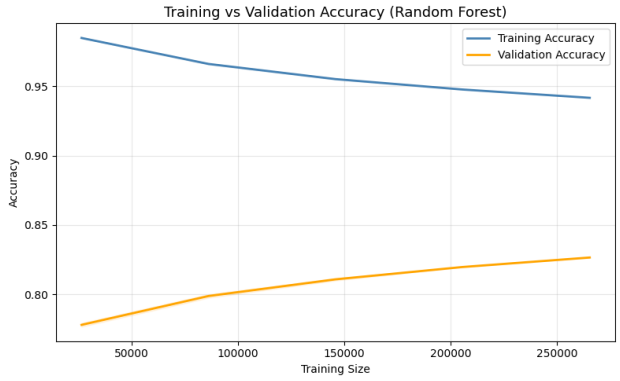


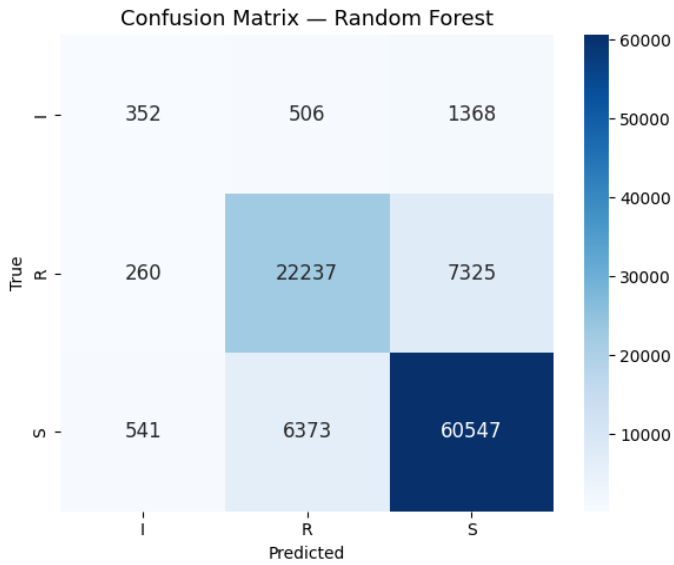
Classification Report:

Accuracy: 77.09%

	precision	recall	f1-score	support
I	0.61	0.01	0.02	2226
R	0.70	0.54	0.61	29822
S	0.79	0.90	0.84	67461
accuracy			0.77	99509
macro avg	0.70	0.48	0.49	99509
weighted avg	0.76	0.77	0.75	99509

Random Forest



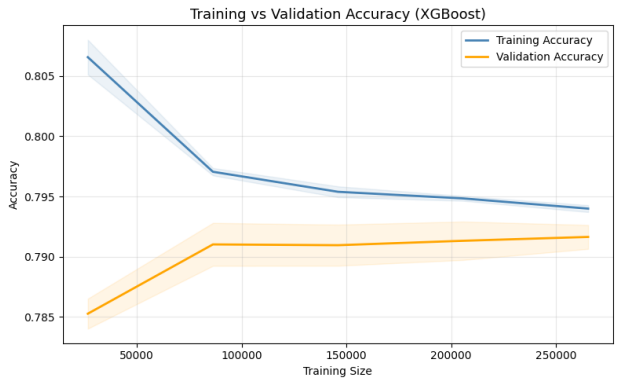


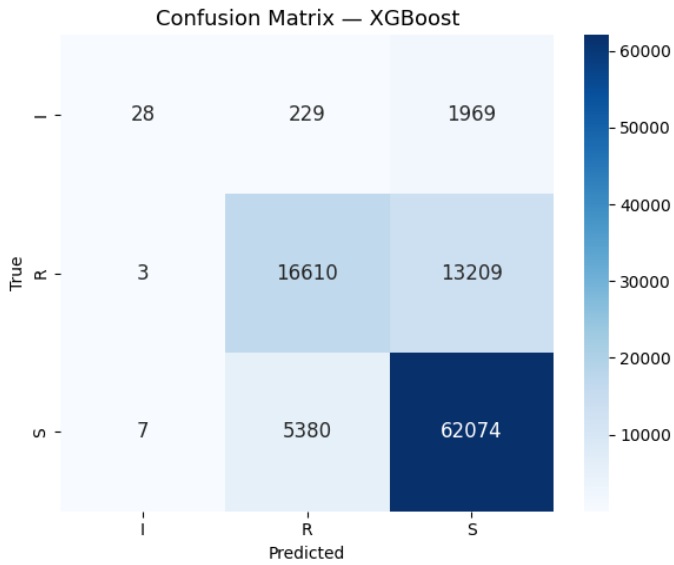
Classification Report:

Accuracy: 83.55%

	precision	recall	f1-score	support
I	0.31	0.16	0.21	2226
R	0.76	0.75	0.75	29822
S	0.87	0.90	0.89	67461
accuracy			0.84	99509
macro avg	0.65	0.60	0.62	99509
weighted avg	0.83	0.84	0.83	99509

XGBoost



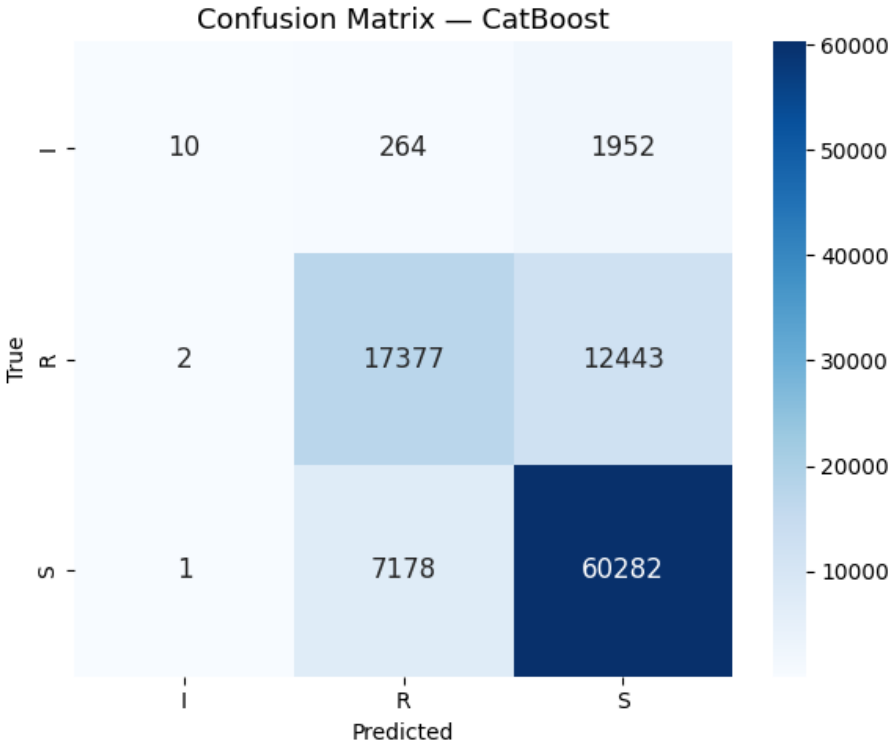


Classification Report:

Accuracy: 79.10%

	precision	recall	f1-score	support
I	0.74	0.01	0.02	2226
R	0.75	0.56	0.64	29822
S	0.80	0.92	0.86	67461
accuracy			0.79	99509
macro avg	0.76	0.50	0.51	99509
weighted avg	0.79	0.79	0.77	99509

CatBoost



Classification Report:

Accuracy: 78.05%

	precision	recall	f1-score	support
I	0.77	0.00	0.01	2226
R	0.70	0.58	0.64	29822
S	0.81	0.89	0.85	67461
accuracy			0.78	99509
macro avg	0.76	0.49	0.50	99509
weighted avg	0.77	0.78	0.77	99509

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STEP 6.7 — Model Comparison

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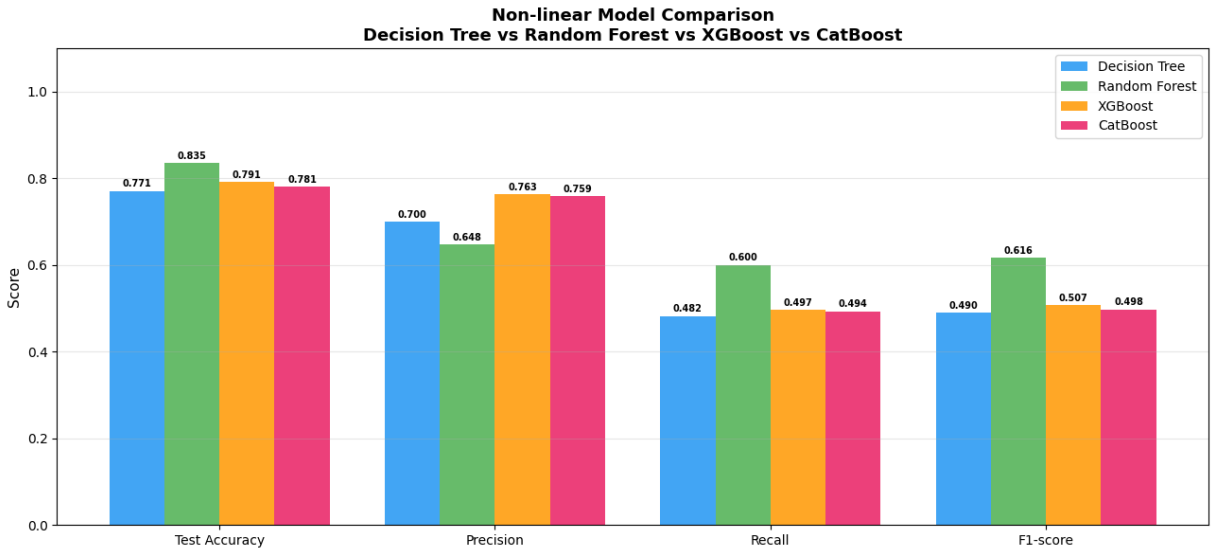
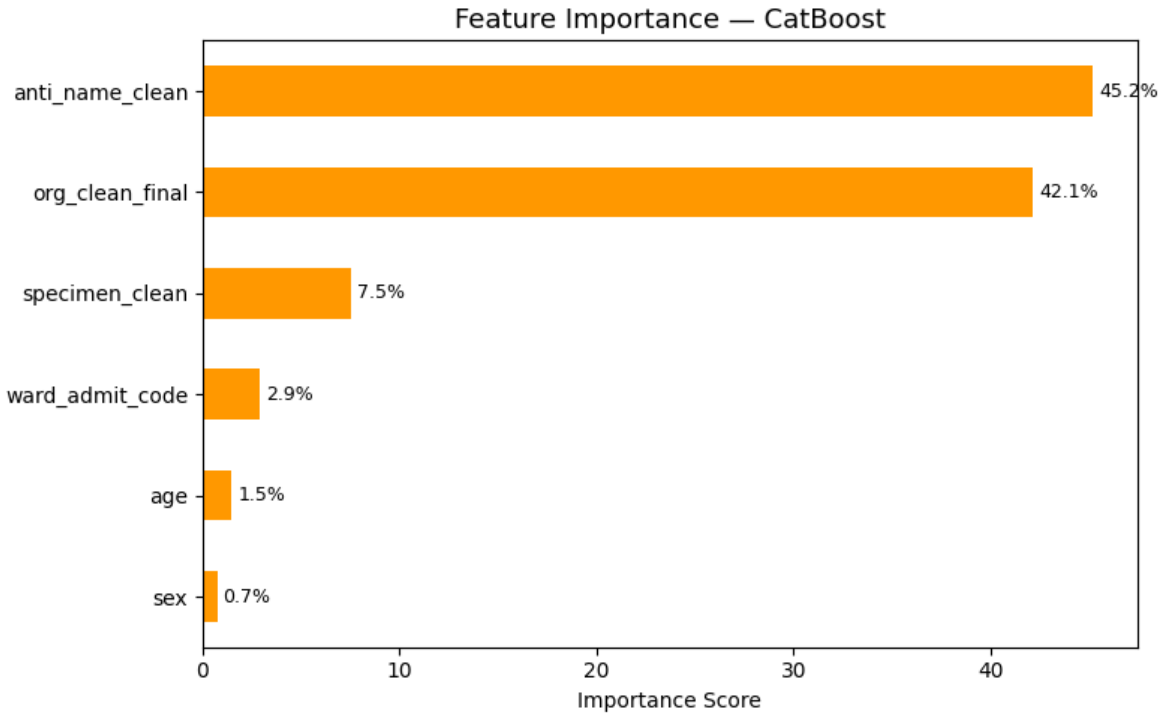
Model	Test Acc	Precision	Recall	F1
Decision Tree	0.7709	0.6997	0.4822	0.4902
Random Forest	0.8355	0.6478	0.6004	0.6163 ✅
XGBoost	0.7910	0.7626	0.4966	0.5070
CatBoost	0.7805	0.7589	0.4936	0.4977





Best Model : Random Forest

Test Accuracy : 83.55%

Macro F1 : 0.6163

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